DEPARTMENT OF INFORMATION TECHNOLOGY
LAB MANUAL

CS23432 – Software Construction

(REGULATION 2023)

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Year / Branch / Section: II / B.Tech IT

Semester: IV

Academic Year: 2025-2026

INDEX

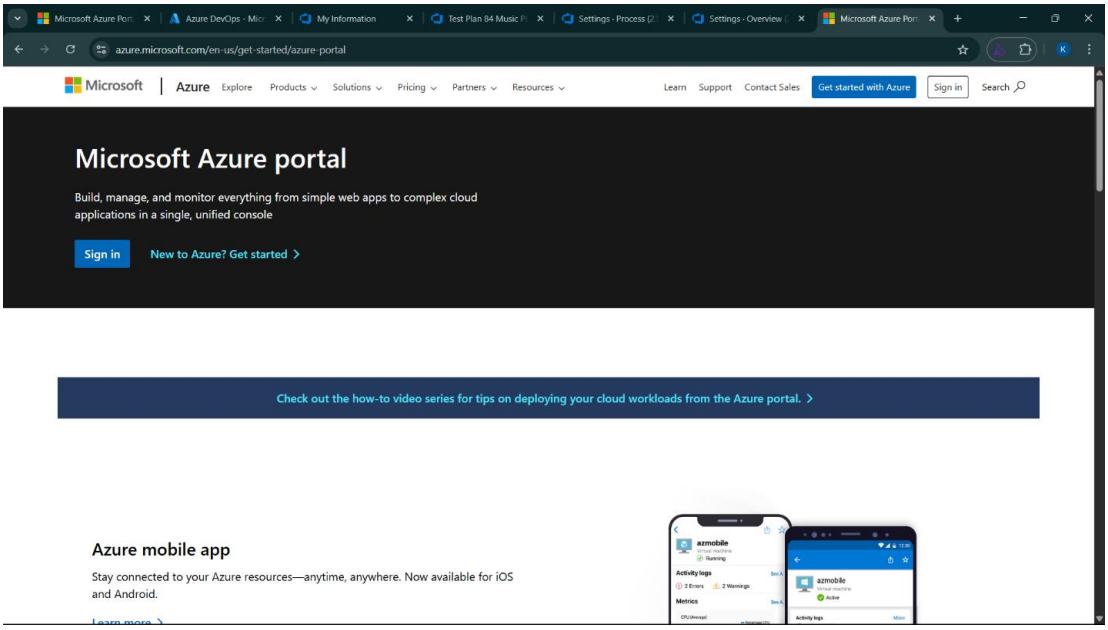
EXP NO: 1	AZURE DEVOPS ENVIRONMENT SETUP

Aim:

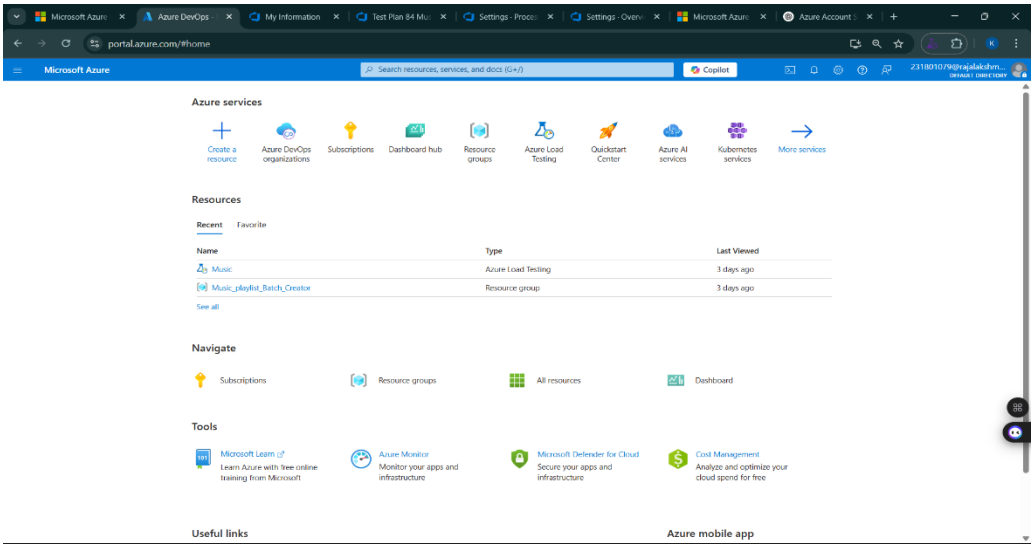
To set up and access the Azure DevOps environment by creating an organization through the Azure portal.

INSTALLATION

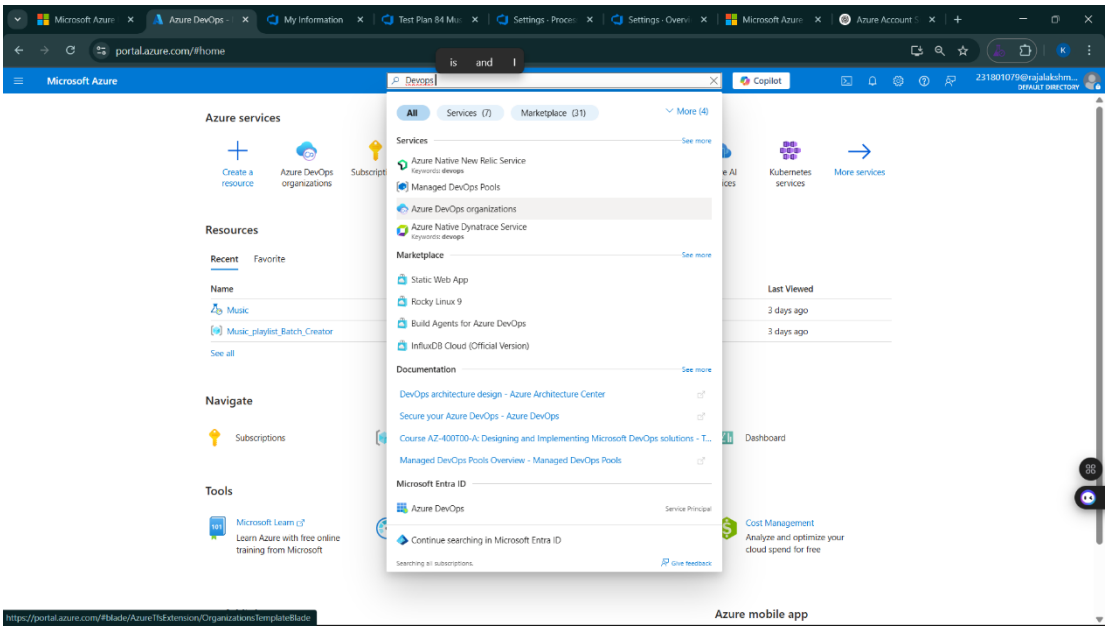
1.Open your web browser and go to the Azure website: https://azure.microsoft.com/en-us/get-started/azure-portal.
Sign in using your Microsoft account credentials.
If you don't have a Microsoft account, you can create one here: https://signup.live.com/?lic=1



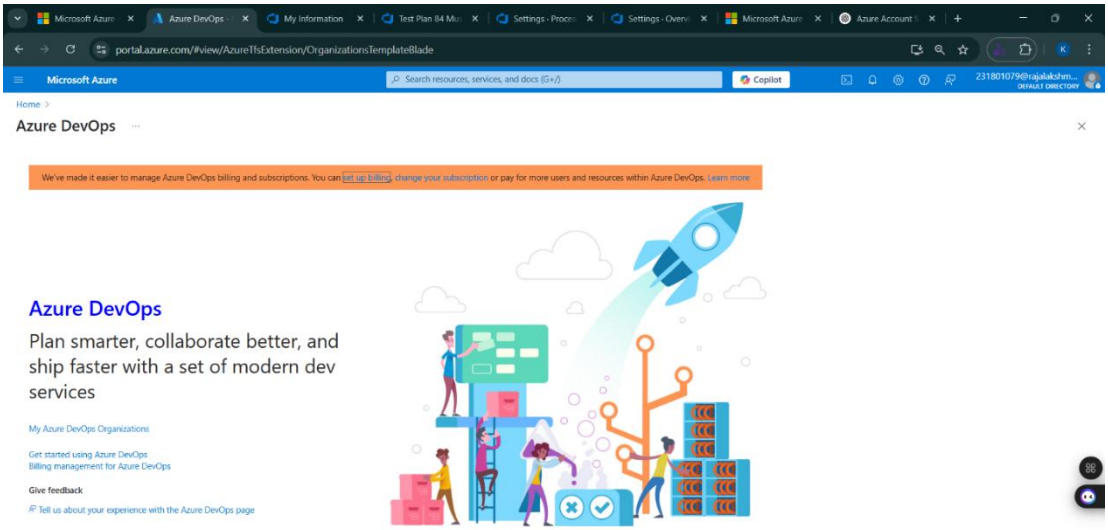
2.Azure home page



3.Open DevOps environment in the Azure platform by typing ***Azure DevOps Organizations*** in the search bar.



4.Click on the ***My Azure DevOps Organization*** link and create an organization and you should be taken to the Azure DevOps Organization Home page.



Result:

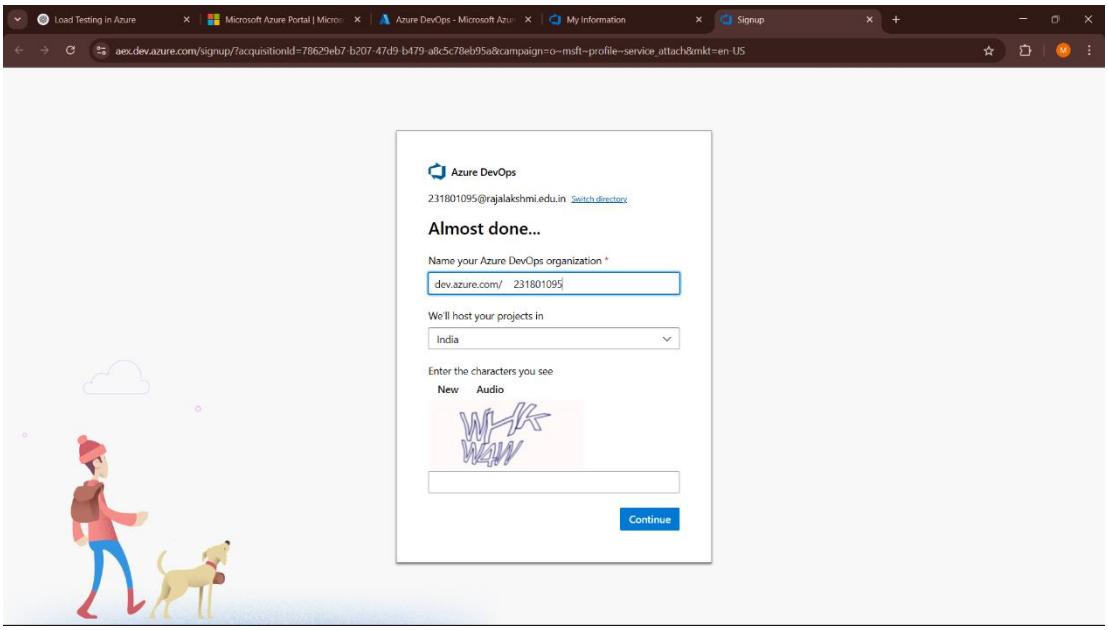
Successfully accessed the Azure DevOps environment and created a new organization through the Azure portal.

EXP NO: 2	AZURE DEVOPS PROJECT SETUP AND USER STORY MANAGEMENT

Aim:

To set up an Azure DevOps project for efficient collaboration and agile work management.

1.Create An Azure Account



2.Create the First Project in Your Organization

a. After the organization is set up, you'll need to create your first **project**. This is where you'll begin to manage code, pipelines, work items, and more.

b. On the organization's **Home page**, click on the **New Project** button.

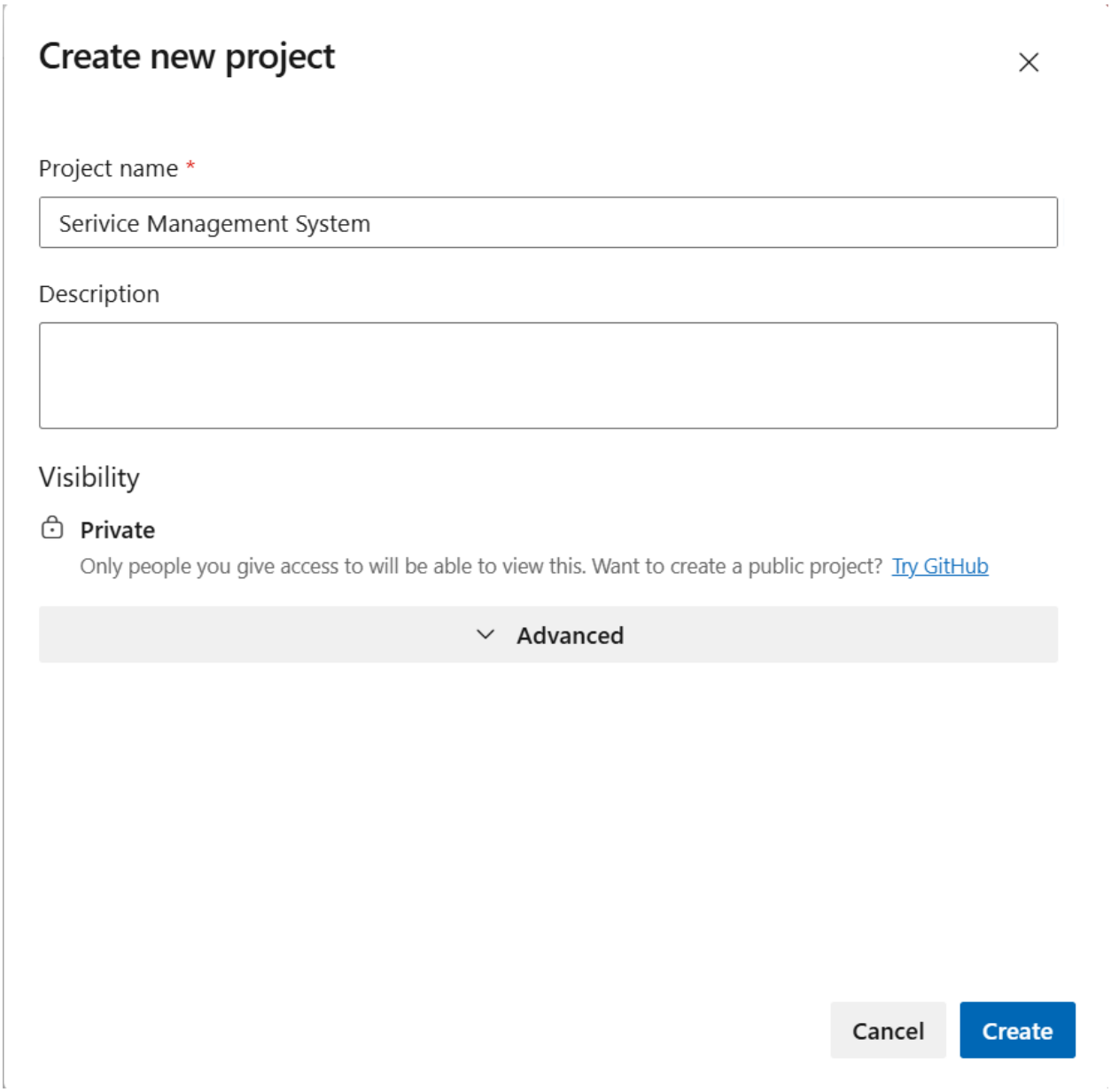
c. Enter the project name, description, and visibility options:

***Name**:* Choose a name for the project (e.g., LMS).

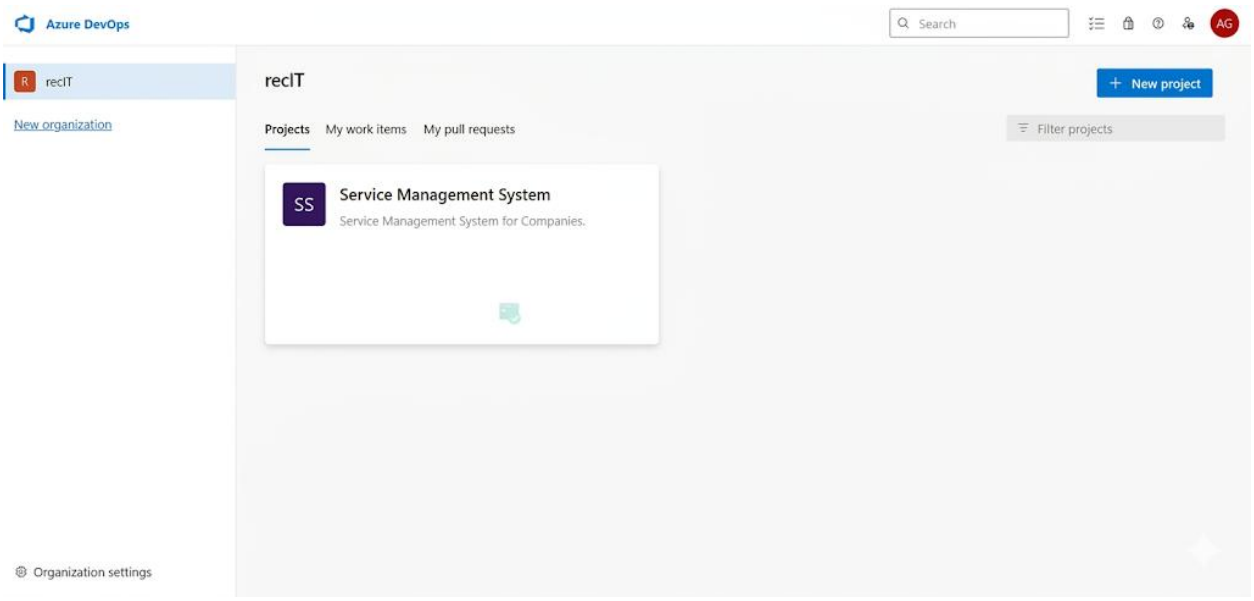
***Description**:* Optionally, add a description to provide more context about the project.

***Visibility**:* Choose whether you want the project to be **Private** (accessible only to those invited) or **Public** (accessible to anyone).

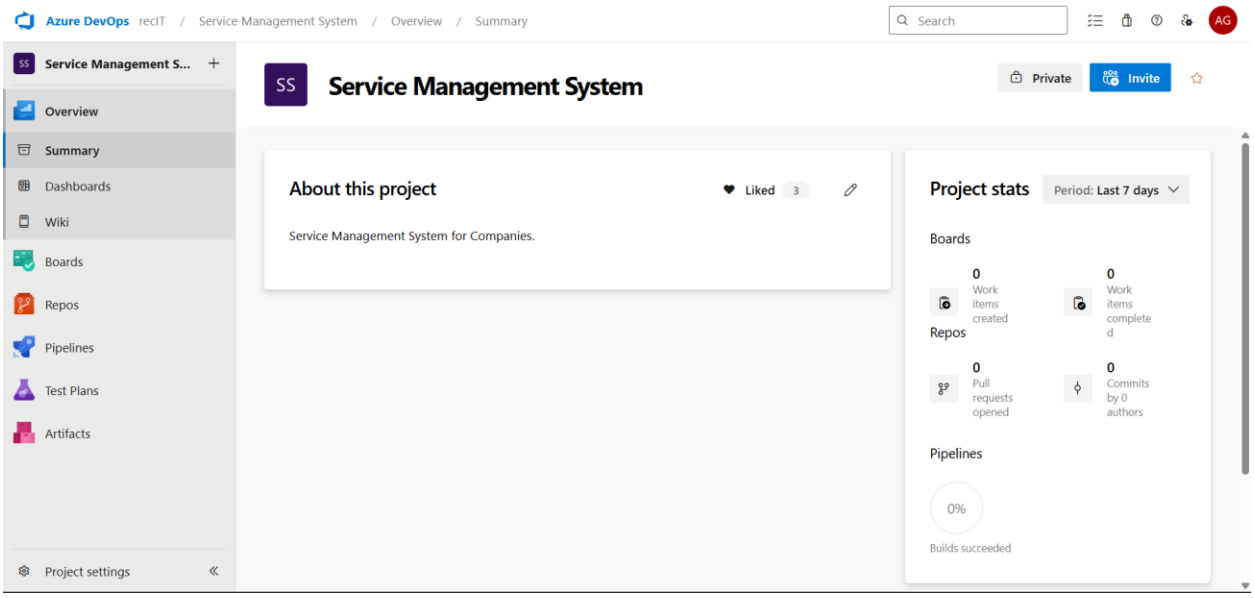
d. Once you've filled out the details, click **Create** to set up your first project.



3.Once logged in, ensure you are in the correct organization. If you're part of multiple organizations, you can switch between them from the top left corner (next to your user profile). Click on the Organization name, and you should be taken to the Azure DevOps Organization Home page.



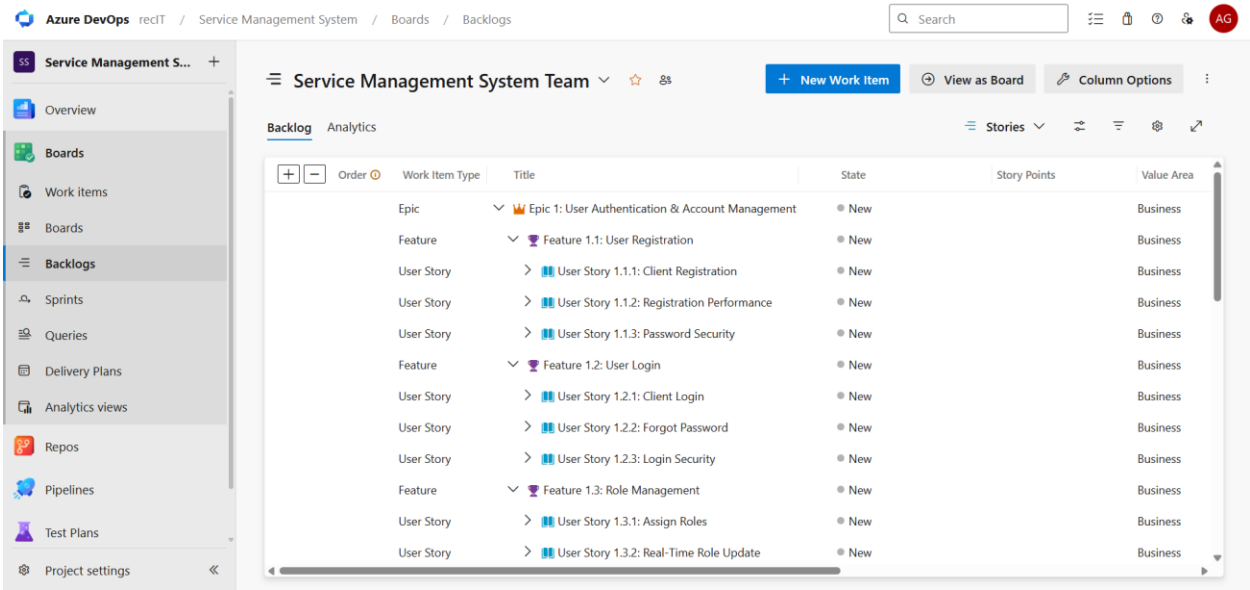
4.Project dashboard



5.To manage user stories:

a. From the **left-hand navigation menu**, click on **Boards**. This will take you to the main **Boards** page, where you can manage work items, backlogs, and sprints.

b. On the **work items** page, you'll see the option to **Add a work item** at the top. Alternatively, you can find a + button or **Add New Work Item** depending on the view you're in. From the **Add a work item** dropdown, select **User Story**. This will open a form to enter details for the new User Story.



Result:

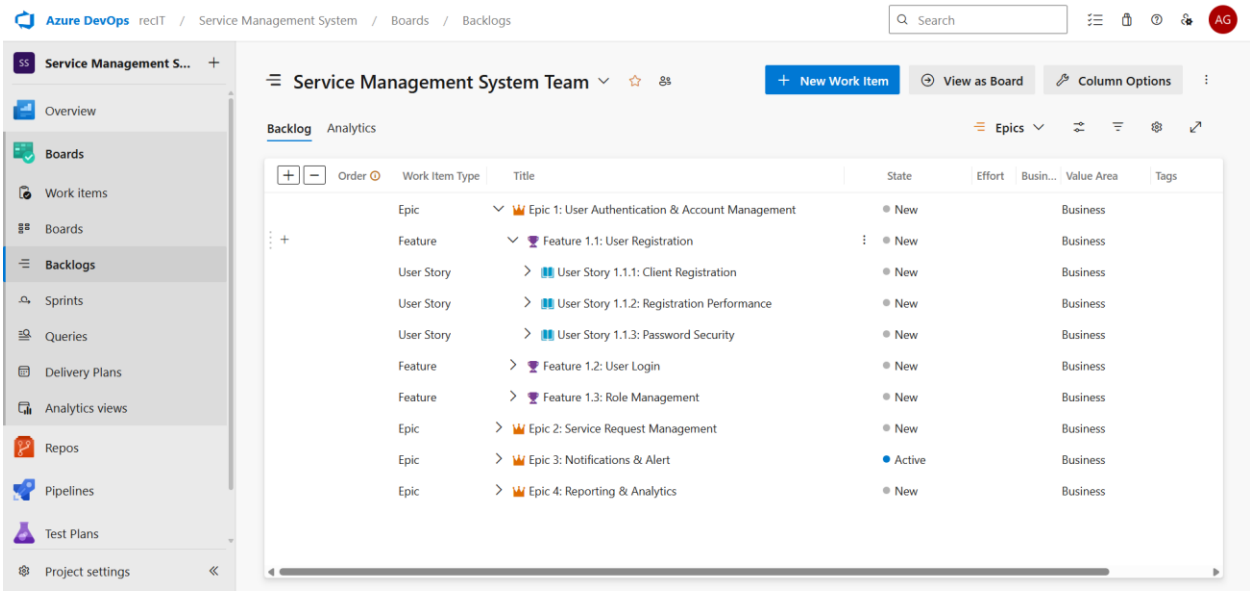
Successfully created an Azure DevOps project with user story management and agile workflow setup.

EXP NO: 3	SETTING UP EPICS, FEATURES, AND USER STORIES FOR PROJECT PLANNING

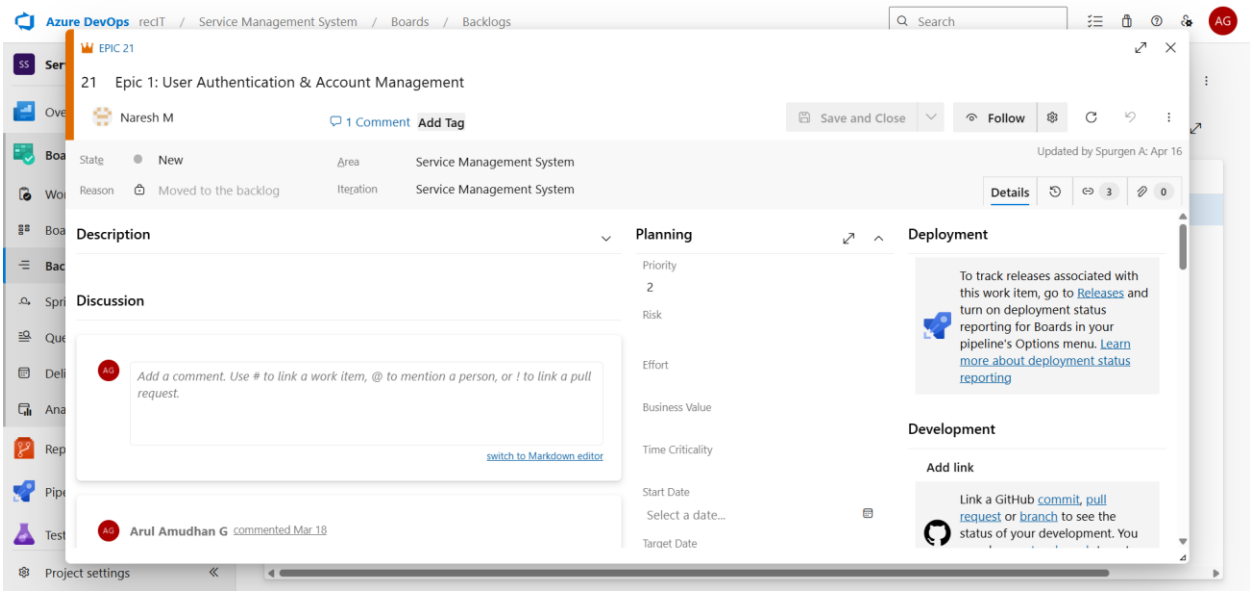
Aim:

To learn about how to create epics, user story, features, backlogs for your assigned project.

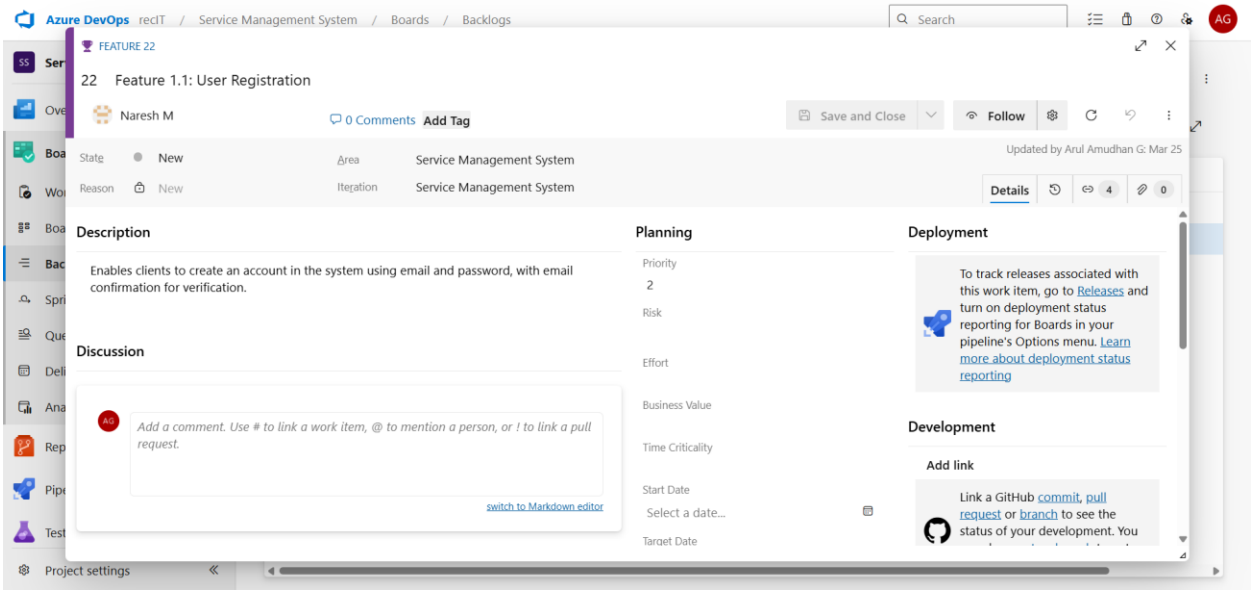
Create Epic, Features, User Stories, Task



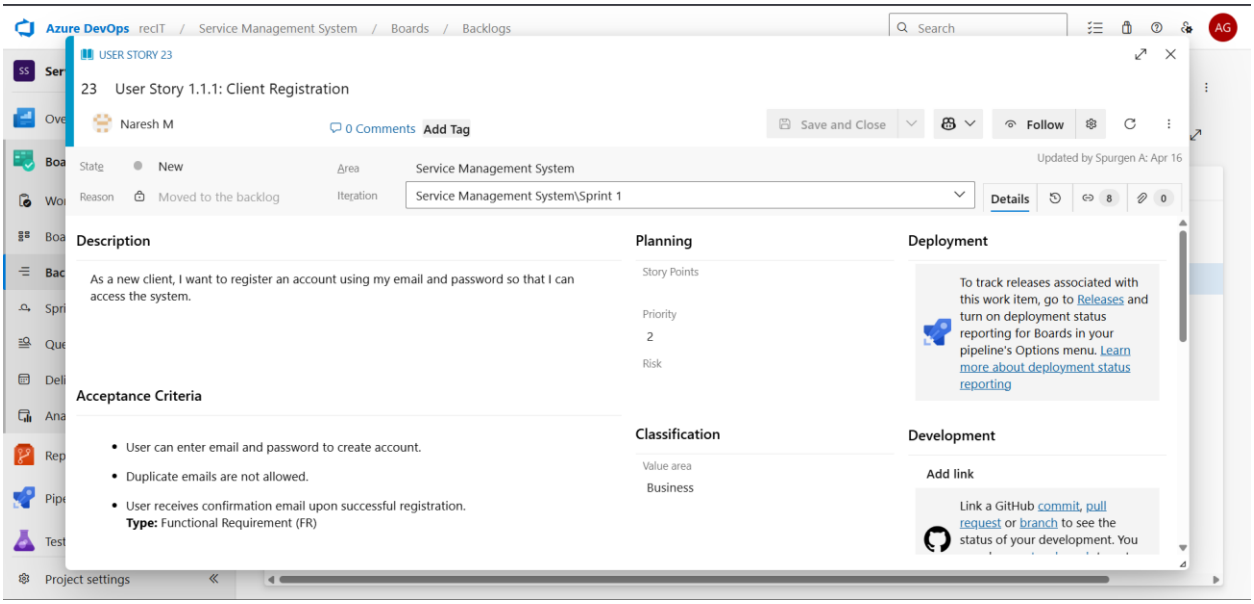
1.Fill in Epics



2.Fill in Features



3.Fill in User Story Details



Result:

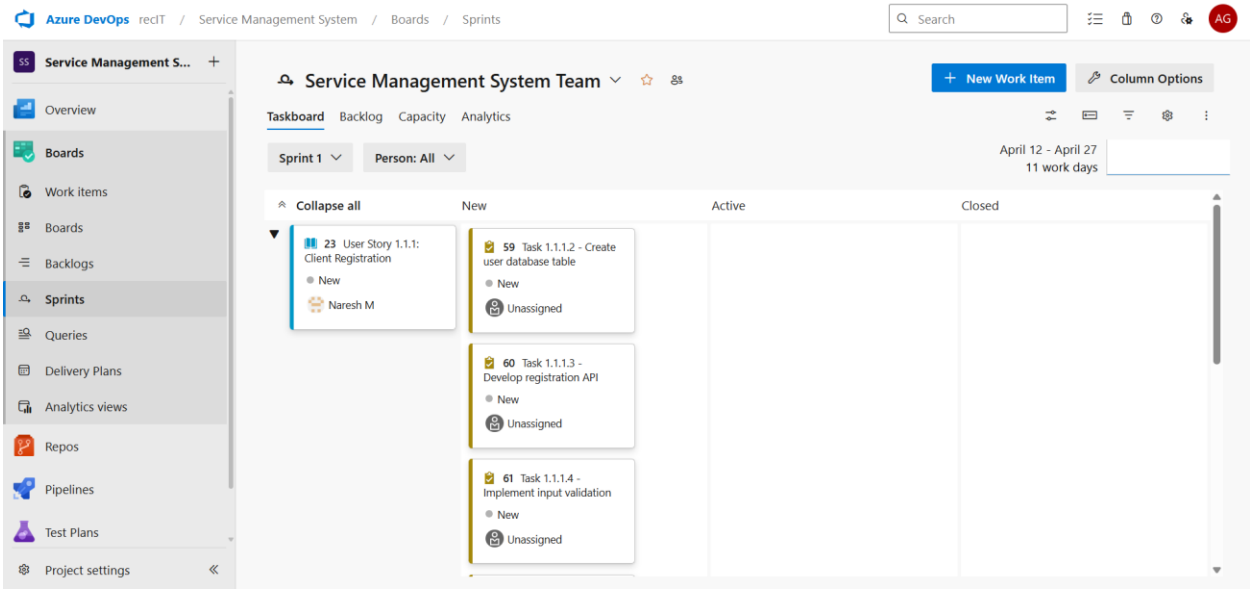
Thus, the creation of epics, features, user story and task has been created successfully.

EXP NO: 4	SPRINT PLANNING

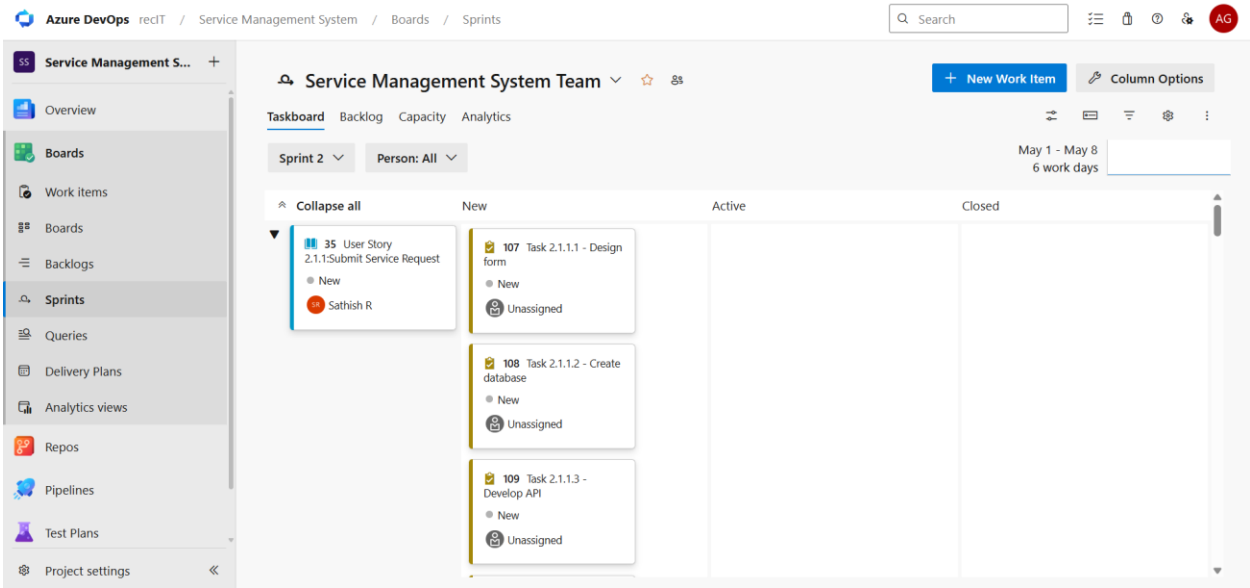
Aim:

To assign user story to specific sprint for the Service Management System Project.

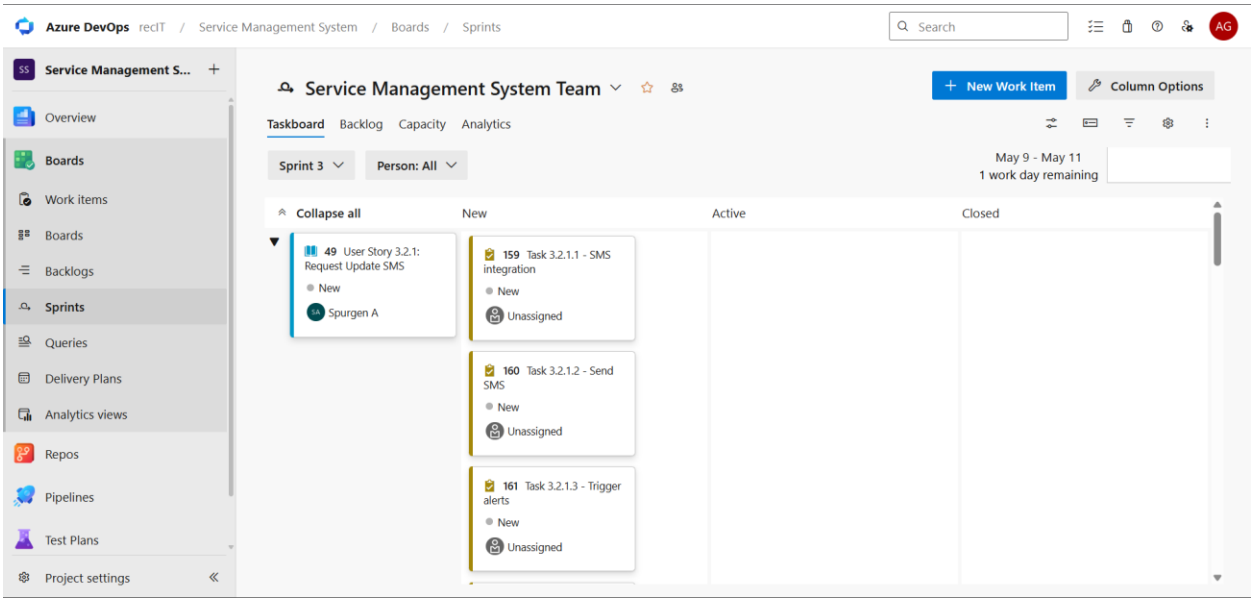
Sprint Planning
Sprint 1



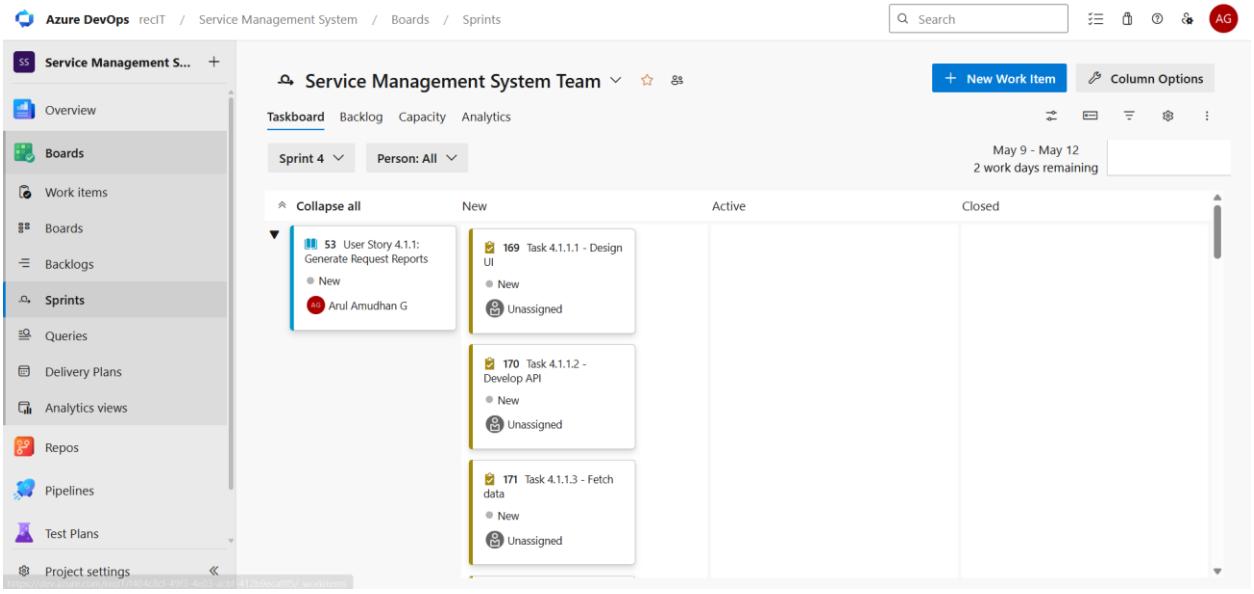
Sprint 2



Sprint 3



Sprint 4



Result:

The Sprints are created for the Service Management System Project.

EXP NO: 5	POKER ESTIMATION

Aim:

Create Poker Estimation for the user stories - Service Management System Project.

Planning Poker Estimation:

User Story 1: Submit Service Request
User Story ID: 01

Title: Submit Service Request

Description: As a Client, I want to submit a service request with details like service type, description, and priority so that the administration can review and assign my request.

Story Point Estimation: 5 Points

Team Poker Votes:

Team Member	Vote (Story Point)
Team Member 1	3
Team Member 2	5
Team Member 3	5
Team Member 4	8

Final Consensus: 5 Points

Reason for Estimation:

- Requires building a dynamic frontend form with validation.
- Backend logic involves mapping the request to the currently authenticated user (req.user.id).
- Database integration with the ServiceRequest model using Mongoose.
- Includes handling different priority levels and service types.

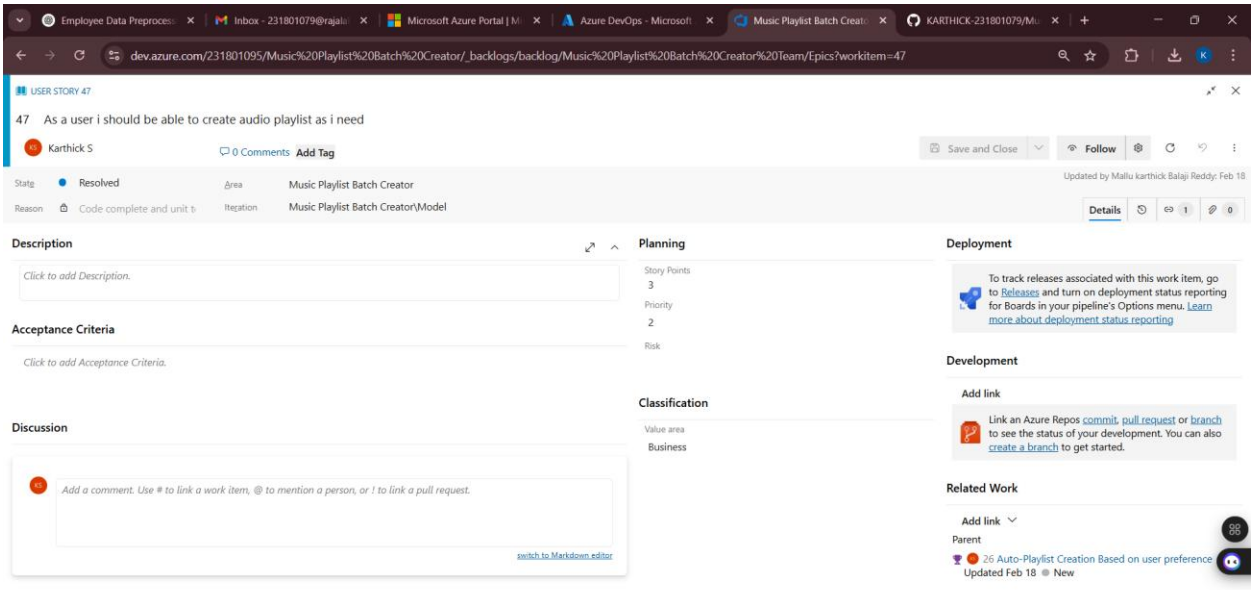
Risks:

- Data validation failures (e.g., empty descriptions).
- Inconsistent priority handling between UI and Backend.

Acceptance Criteria Covered:

- Client can enter serviceType, description, and priority.

- System automatically assigns "Pending Approval" status to new requests.
- Request is correctly linked to the logged-in client’s ID.
- Successful submission redirects the user to their dashboard.



Result:

The Estimation/Story Points is created for the project using Poker Estimation.

EXP NO: 6	DESIGNING CLASS AND SEQUENCE DIAGRAMS FOR PROJECT ARCHITECTURE

Aim:

To Design a Class Diagram and Sequence Diagram for the given Project.

6A. Class Diagram

Classes:

- User
- ServiceRequest
- Feedback
- Ticket
- ChatMessage

Key Attributes:

User: name, email, password, role (Client, Admin, Developer, Tester), isVerified

ServiceRequest: serviceType, description, priority, status, devStatus, testerStatus, bugsReported, slaDeadline

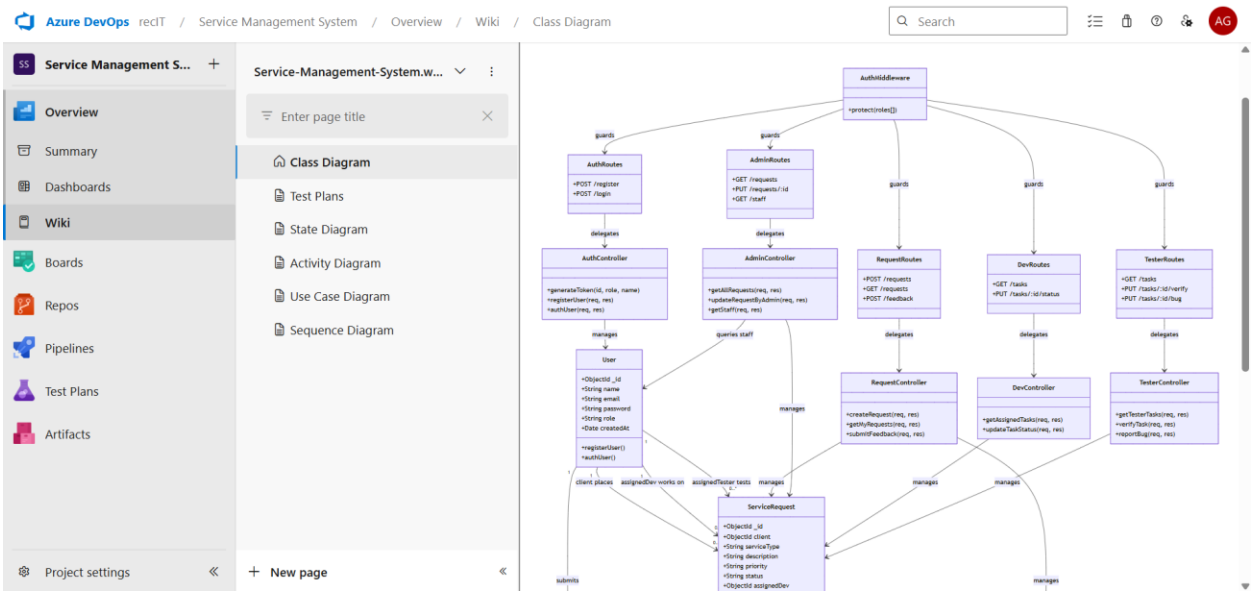
Feedback: rating, comments, createdAt

Ticket: subject, description, status, priority, createdAt

ChatMessage: message, read, createdAt

Relationships:

- User (Client) → creates → ServiceRequest
- User (Admin) → assigns → ServiceRequest
- User (Developer) → works on → ServiceRequest
- User (Tester) → verifies → ServiceRequest
- User (Client) → submits → Feedback
- Feedback → belongs to → ServiceRequest
- User → opens → Ticket
- User → sends → ChatMessage
- ChatMessage → linked to → ServiceRequest



6B. Sequence Diagram

Actors:

Client / Admin / Developer / Tester → Frontend → Server → Database (DB)

Steps:

1. Client enters registration details
2. Frontend sends register request to Server
3. Server saves user in DB
4. Server returns JWT token to Client
5. Client enters login credentials
6. Frontend sends login request to Server
7. Server verifies credentials in DB
8. Server returns JWT token to Client
9. Client submits a new service request
10. Frontend sends service request with token to Server
11. Server saves service request in DB
12. Server returns success response to Client
13. Admin requests dashboard data
14. Server fetches all requests and staff list from DB
15. Server returns requests and staff list to Admin
16. Admin assigns a Developer and Tester to the request
17. Server updates the request assignments in DB
18. Developer requests assigned tasks
19. Server fetches assigned tasks from DB and returns to Developer
20. Developer marks task as "Completed"
21. Server updates devStatus and changes overall status to "In Testing" in DB
22. Tester requests ready-for-testing tasks
23. Server fetches tasks from DB and returns to Tester

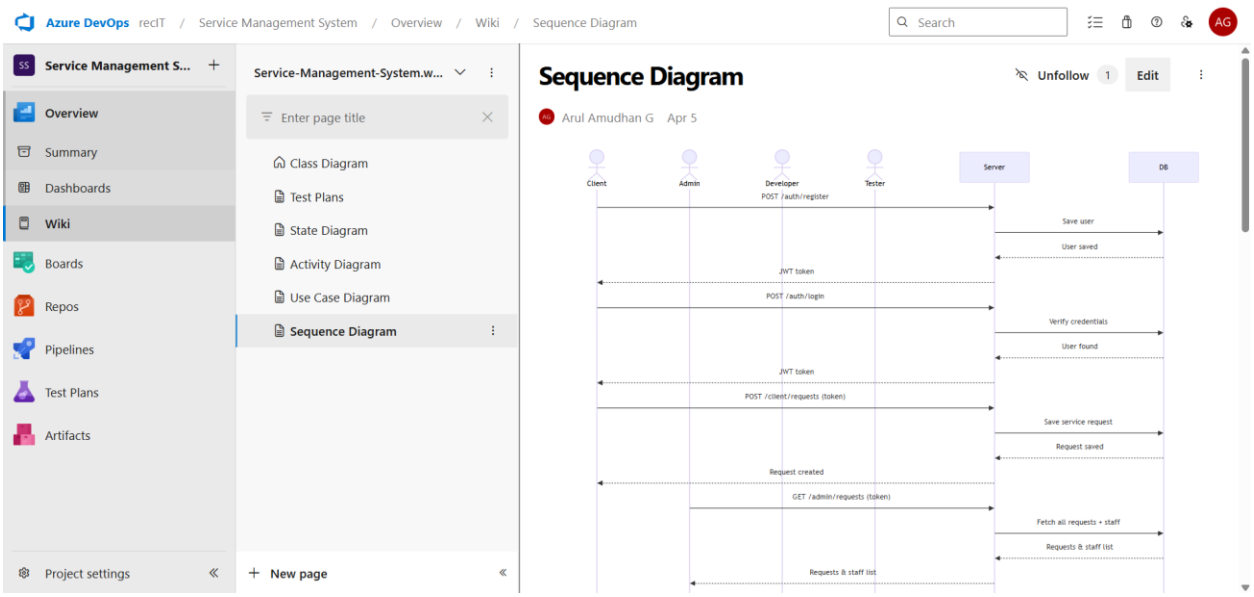
24. Tester evaluates the completed task
25. Admin marks the final request status as "Completed"
26. Server updates final status in DB
27. Client submits feedback for the completed request
28. Server saves feedback in DB
29. Service request lifecycle successful

Condition: Bugs Found

• Tester sends bug report to Server
• Server updates status to "In Development" in DB
• Task is sent back to Developer for fixes

Condition: Verified Successfully

• Tester sends verify request to Server
• Server updates status to "Ready for Delivery" in DB



Result:

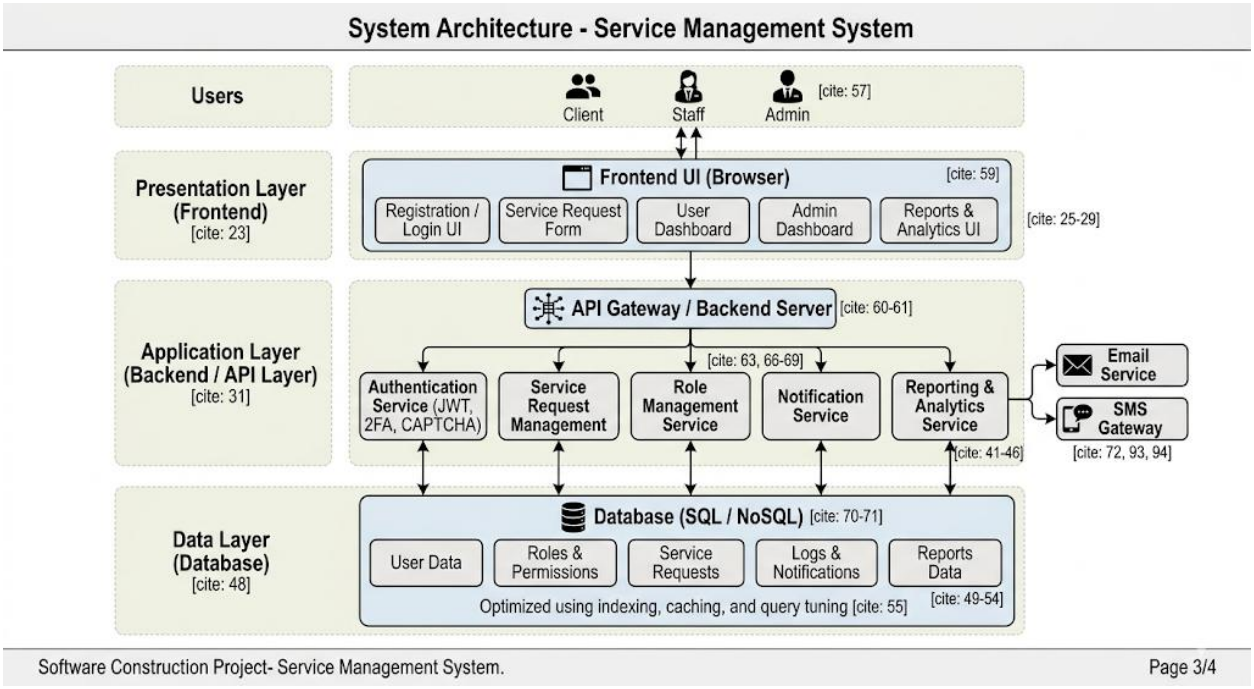
The Class Diagram and Sequence Diagram is designed Successfully for Service Management System Project.

EXP NO: 7	DESIGNING ARCHITECTURAL AND ER DIAGRAMS FOR PROJECT STRUCTURE

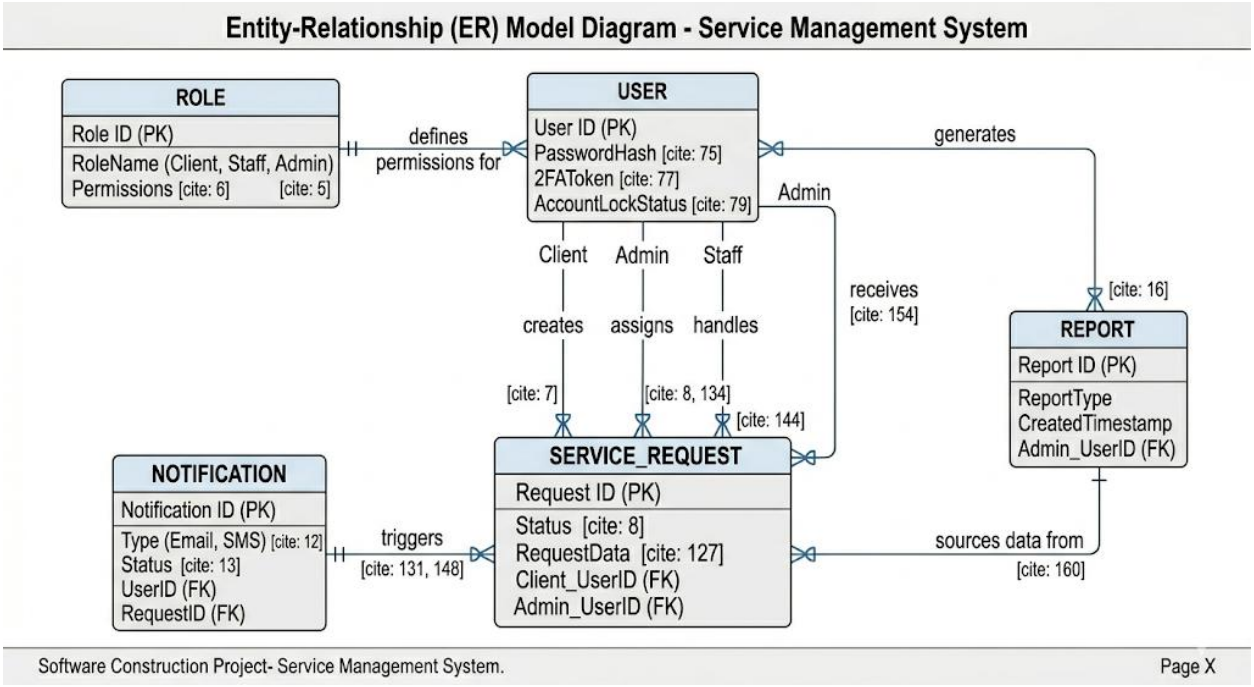
Aim:

To Design an Architectural Diagram and ER Diagram for the given Project.

7A. Architectural Diagram:



7B.ER Diagram:



Result:

The Architecture Diagram and ER Diagram is designed Successfully for the Service Management System.

EXP NO: 8	TESTING – TEST PLANS AND TEST CASES

Aim:

Test Plans and Test Case and write two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.

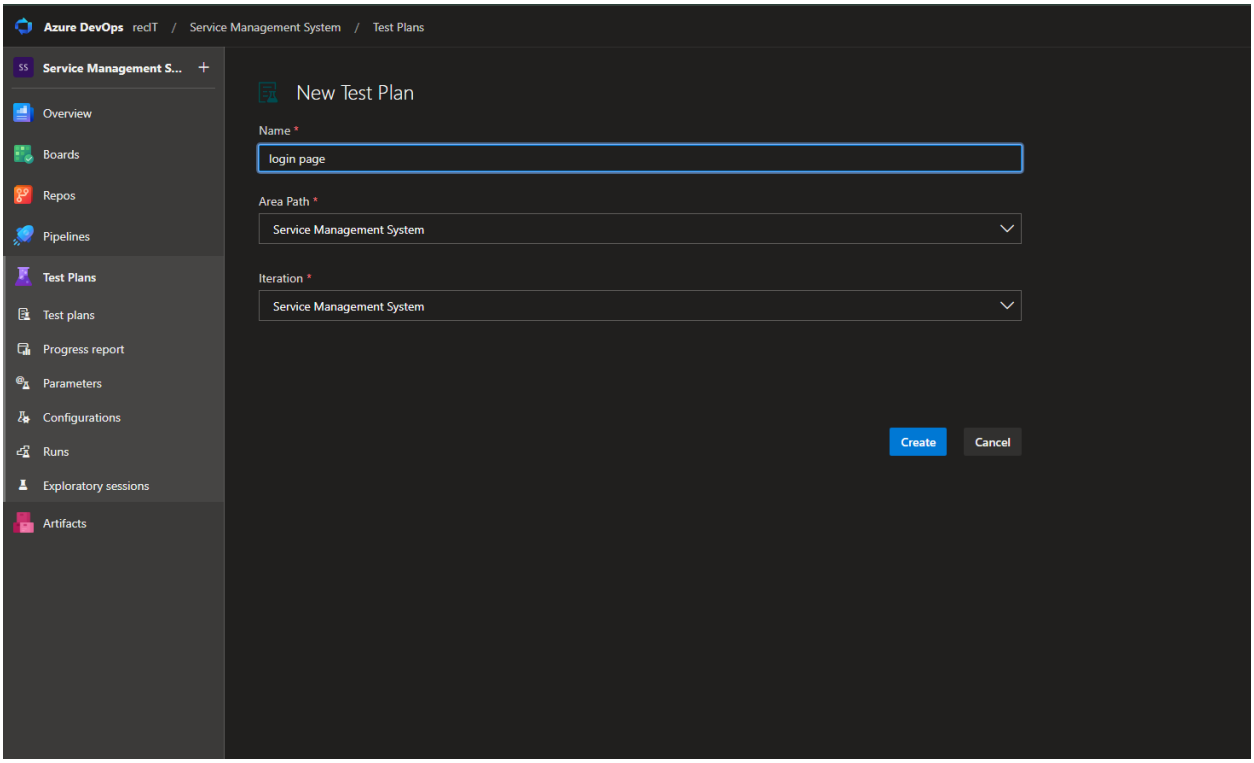
Test Case Design Procedure

Procedure

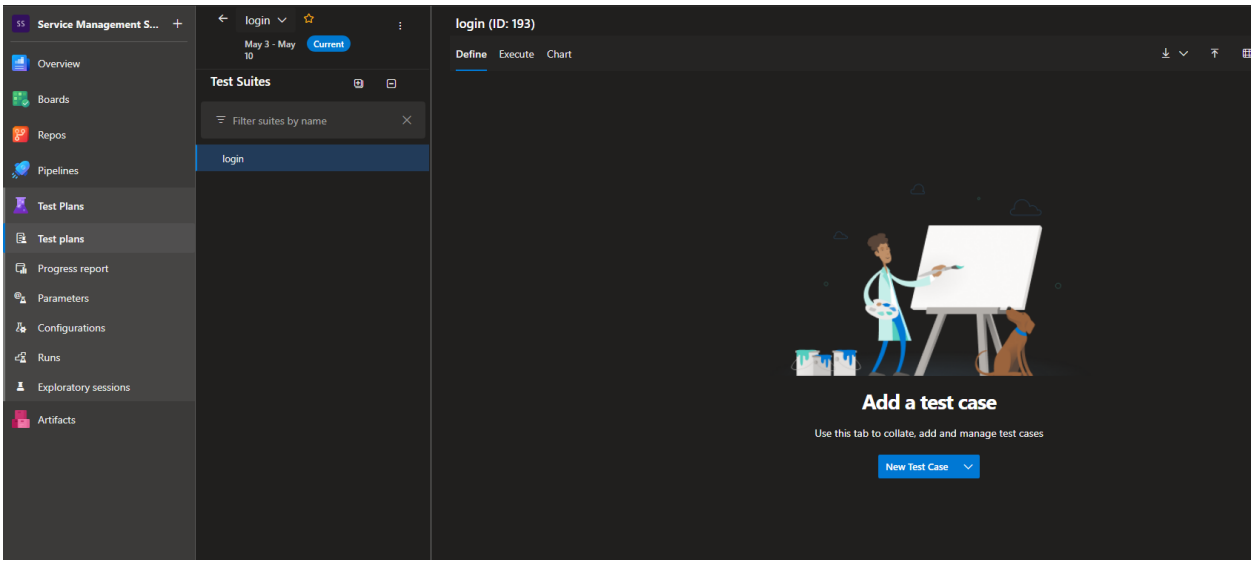
1. Identify system modules such as Login, Registration, Request Management, and Admin Control.
2. Define input values and expected outputs for each module.
3. Create positive and negative test cases.
4. Execute the test cases manually or using testing tools.
5. Compare actual output with expected results.
6. Record defects and update the system if errors occur.

1. **Use Clear Naming and IDs**
    - Test cases are named clearly
    - Helps in quick identification and linking to user stories or features.
2. **Separate Test Suites**
    - Grouped test cases based on functionality
    - Improves organization and test execution flow in Azure DevOps.
3. **Prioritize and Review**
    - Critical user actions are marked high-priority.
    - Reviewed for completeness and traceability against feature requirements.

1.New test plan



2.Test suite



3.Test case

Give two test cases for at least five user stories showcasing the happy path and error scenarios in azure DevOps platform.
Service Management System – Test Plans

USER STORIES

1. As a user, I want to sign up and log in securely so that I can access the Service Management System. (ID: 79)
2. As a user, I need to view all my service requests in one place. (ID: 76)
3. As a user, I should be able to create a service request when needed. (ID: 73)
4. As a user, I should be able to edit, update, and change service request details. (ID: 68)
5. As a user, I need to receive real-time notifications and status updates. (ID: 65)

Test Suites

Test Suit: TS01 - User Login (ID: 86)

1. **TC01 – Successful Sign Up**
    - **Action:**
        - Go to the Sign-Up page.
        - Enter valid name, email, and password.
        - Click "Sign Up".
    - **Expected Results:**
        - Sign-Up form is displayed.
        - Fields accept values without error.
        - Account is created, and the user is redirected to the dashboard.
    - **Type**: Happy Path
2. **TC02 – Secure Login**
    - **Action:**
        - Go to the Login page.
        - Enter valid email and password.
        - Click on "Login".
    - **Expected Results:**
        - Login form is displayed.
        - Fields accept data without error.
        - User is logged in and redirected to the dashboard.
    - **Type:** Happy Path
3. **TC03 – Sign Up with Existing Email**
    - **Action:**
        - Go to the Sign-Up page.
        - Enter a name and an already registered email.
        - Click on "Sign Up".
    - **Expected Results:**

        - Fields accept data.
        - Error message "Email already registered" is displayed.
    - **Type:** Error Path

4. **TC04 – Login with Wrong Password**
    - **Action:**
        - Go to the Login page.
        - Enter valid email and incorrect password.
        - Click on "Login".
    - **Expected Results:**
        - Input is accepted.
        - Error message "Invalid username or password" is shown.
    - **Type:** Error Path

Test Suite: TS02 - View Service Requests (ID: 87)

Action:

- Login successfully.
- Navigate to "My Requests".

Expected Results:

- All submitted service requests are displayed clearly.

Type: Happy Path

TC04 – Failed Request Loading

Action:

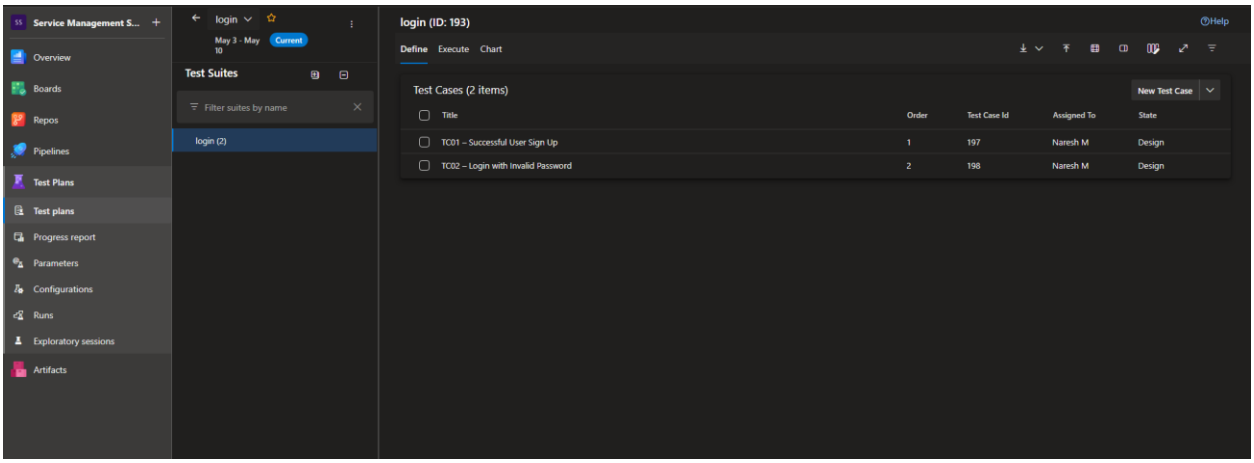
- Disconnect internet connection.
- Open "My Requests".

Expected Results:

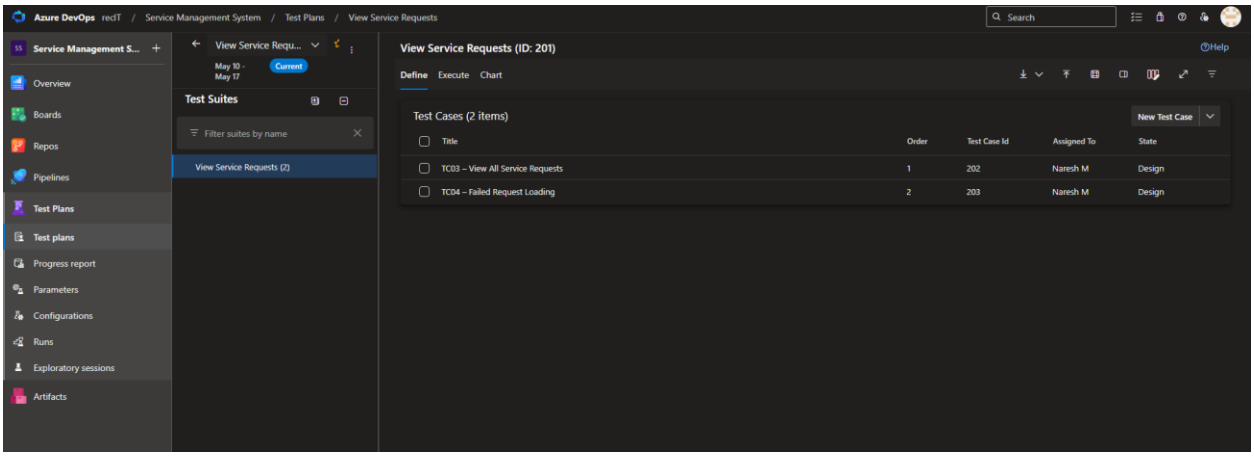
- Requests fail to load.
- Error message "Unable to load requests" is displayed.

Type: Error Path

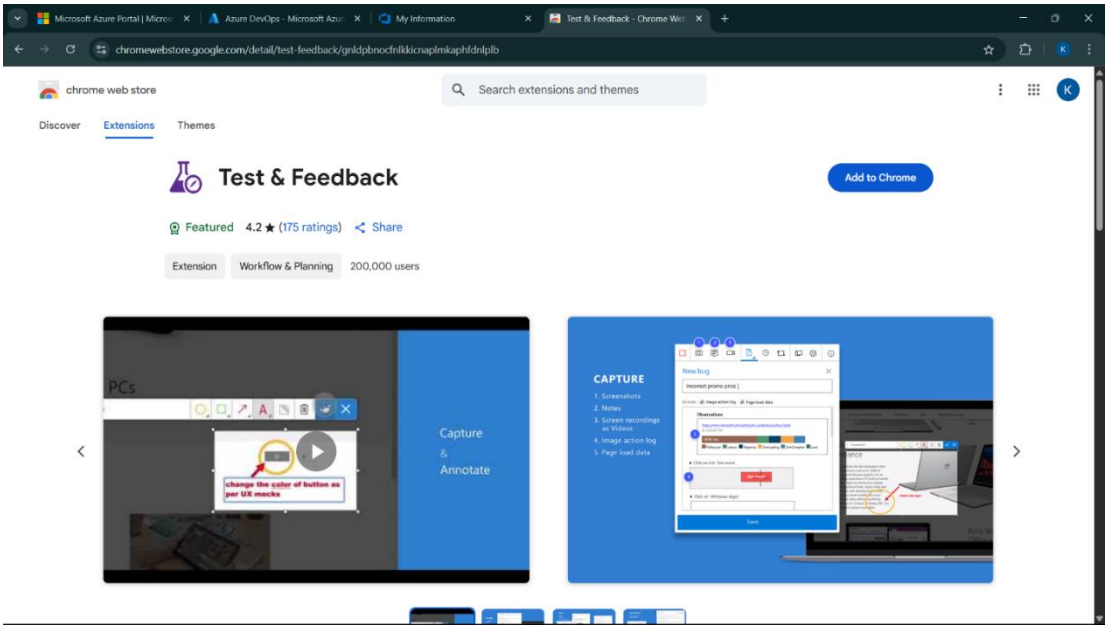
Test Suit 1:



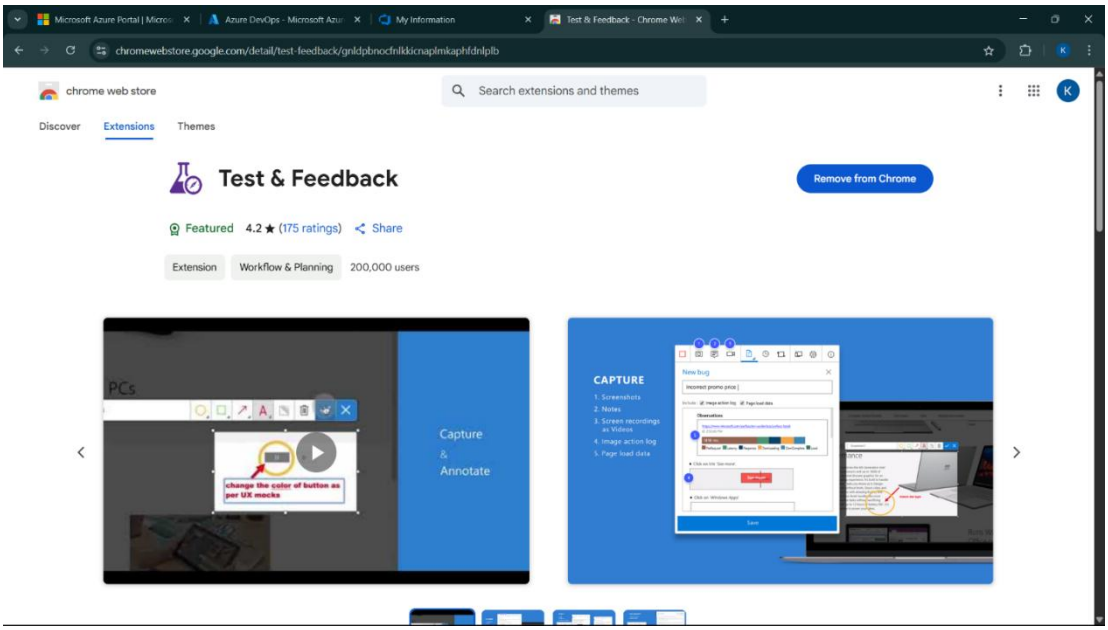
Test Suit 2:



4.Installation of test



5.The test cases are tested successfully and the results are noted.



Result:

The test plans and test cases for the user stories is created in Azure DevOps with Happy Path and Error Path

EXP NO: 9	LOAD TESTING AND PIPELINES

Aim:

To create an Azure Load Testing resource and run a load test to evaluate the performance of a target endpoint and to create and demonstrate an Azure DevOps pipeline for automating application builds, tests, and deployment.

Load Testing

Description:

This experiment demonstrates how load testing is performed for the **Service Management System** to evaluate system performance under multiple user requests. The testing checks response time, request handling, and system stability during heavy traffic conditions.

Steps:

Install Load Testing Tool:

1. Download and install Apache JMeter.
2. Open JMeter and create a new Test Plan.
3. Add Thread Group to simulate multiple users.

Create Load Test Configuration

Configure the following components inside JMeter:

- Thread Group
- HTTP Request Sampler
- Listener for Results
- Summary Report

Load Test Configuration

```
Number of Users (Threads): 50
Ramp-Up Period: 10 Seconds
Loop Count: 5

Server Name: localhost
```

```
Port Number: 5000

Request Type: GET
Path: /api/requests
```

Load Testing Steps

Step 1: Create Thread Group

- Add multiple virtual users.
- Configure ramp-up time and loop count.

Step 2: Add HTTP Request

- Configure API endpoint for testing.
- Select request method such as GET or POST.

Step 3: Add Listener

- Add Summary Report and View Results Tree.
- Monitor request success and failures.

Step 4: Execute Load Test

- Run the test plan.
- Observe response time, throughput, and error rate.

Pipelines:

Description:

This experiment demonstrates how to connect a GitHub-hosted **Service Management System** project with Azure DevOps. The pipeline automatically installs frontend and backend dependencies, builds the application, and publishes artifacts. This helps ensure that every commit triggers automated build validation and smooth deployment.

Steps:

Connect GitHub to Azure DevOps:

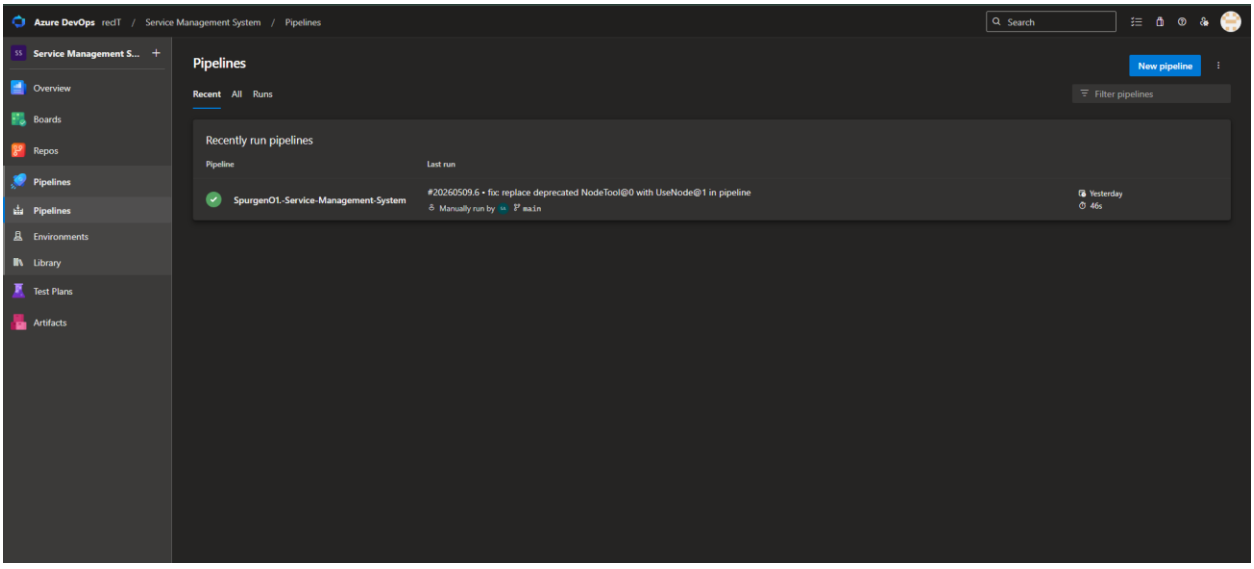
1. Open Azure DevOps and create a new project.

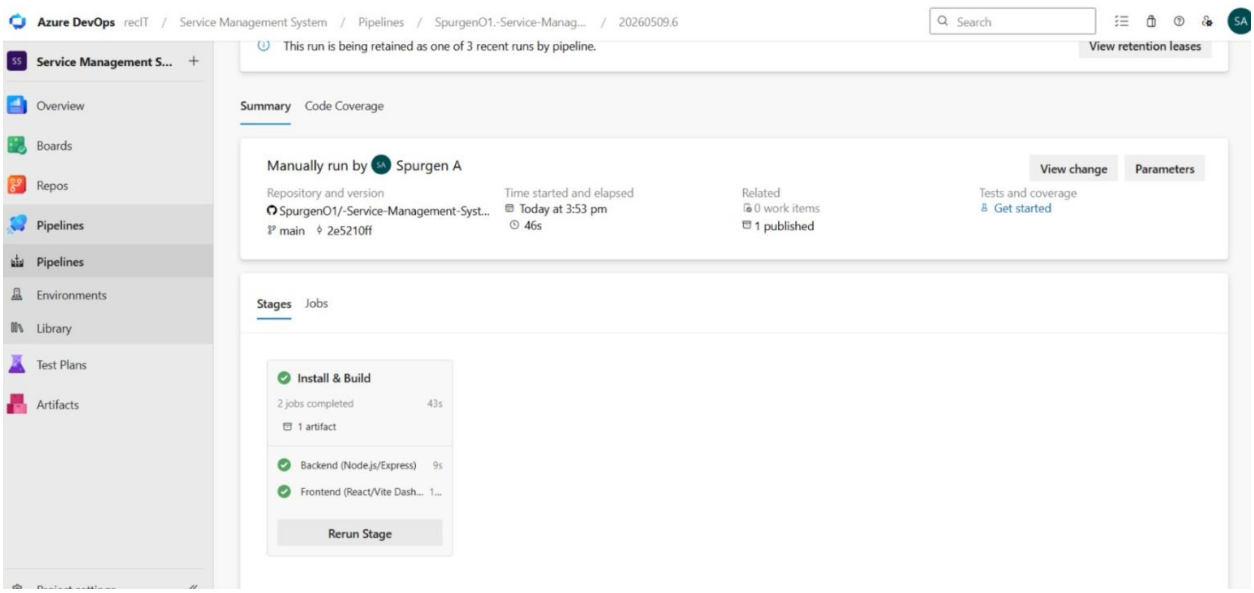
2. Create a new pipeline and select GitHub as the source.
3. Authorize Azure DevOps to access the GitHub repository.
4. Select the Service Management System repository.

Pipeline Tasks Include:

- Connecting GitHub repository with Azure DevOps.
- Installing backend dependencies using npm.
- Building backend services.
- Installing frontend dependencies.
- Building the React/Vite frontend application.
- Running verification steps to ensure successful execution.
- Publishing build artifacts for deployment.

Pipelines:





Result:

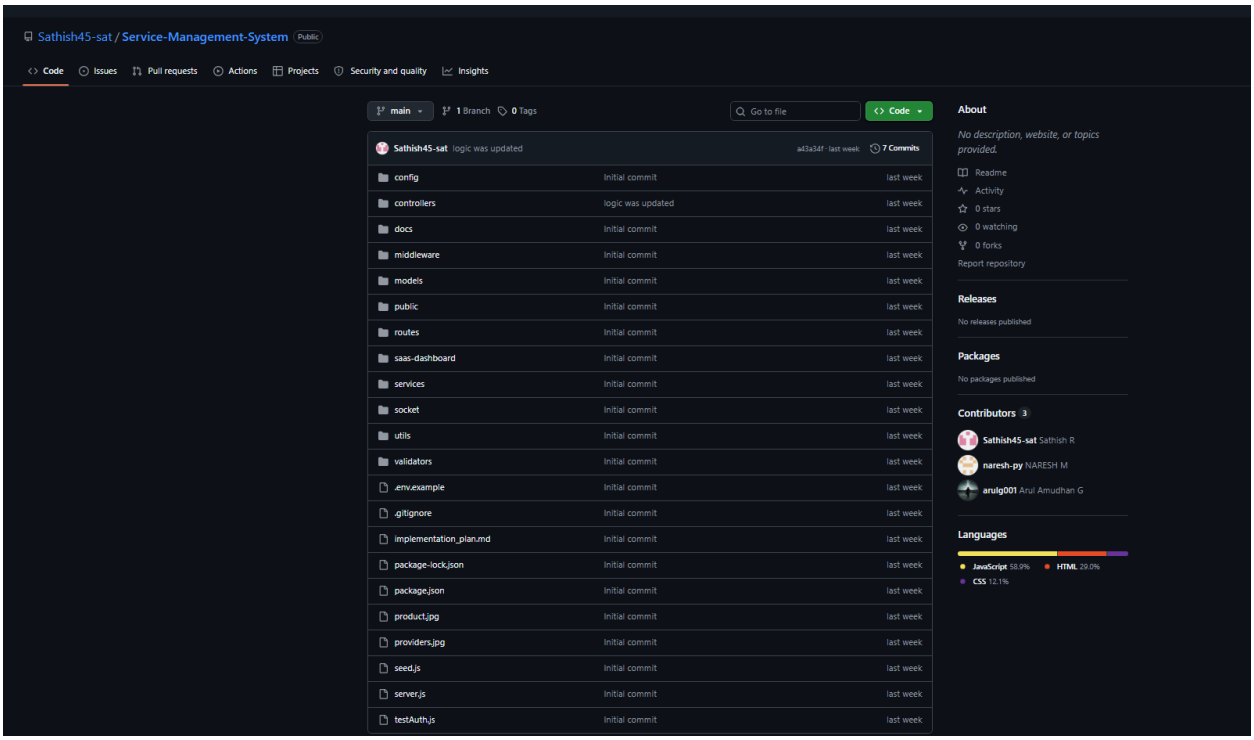
Successfully created the Azure Load Testing resource and executed a load test to assess the performance of the specified endpoint and also demonstrated pipelines in azure devops.

EXP NO: 10	GITHUB: PROJECT STRUCTURE & NAMING CONVENTIONS

Aim:

To provide a clear and organized view of the project's folder structure and file naming conventions, helping contributors and users easily understand, navigate, and extend the Music Playlist Batch Creator project.

GitHub Project Structure



Result:

The GitHub repository clearly displays the organized project structure and consistent naming conventions, making it easy for users and contributors to understand and navigate the codebase.